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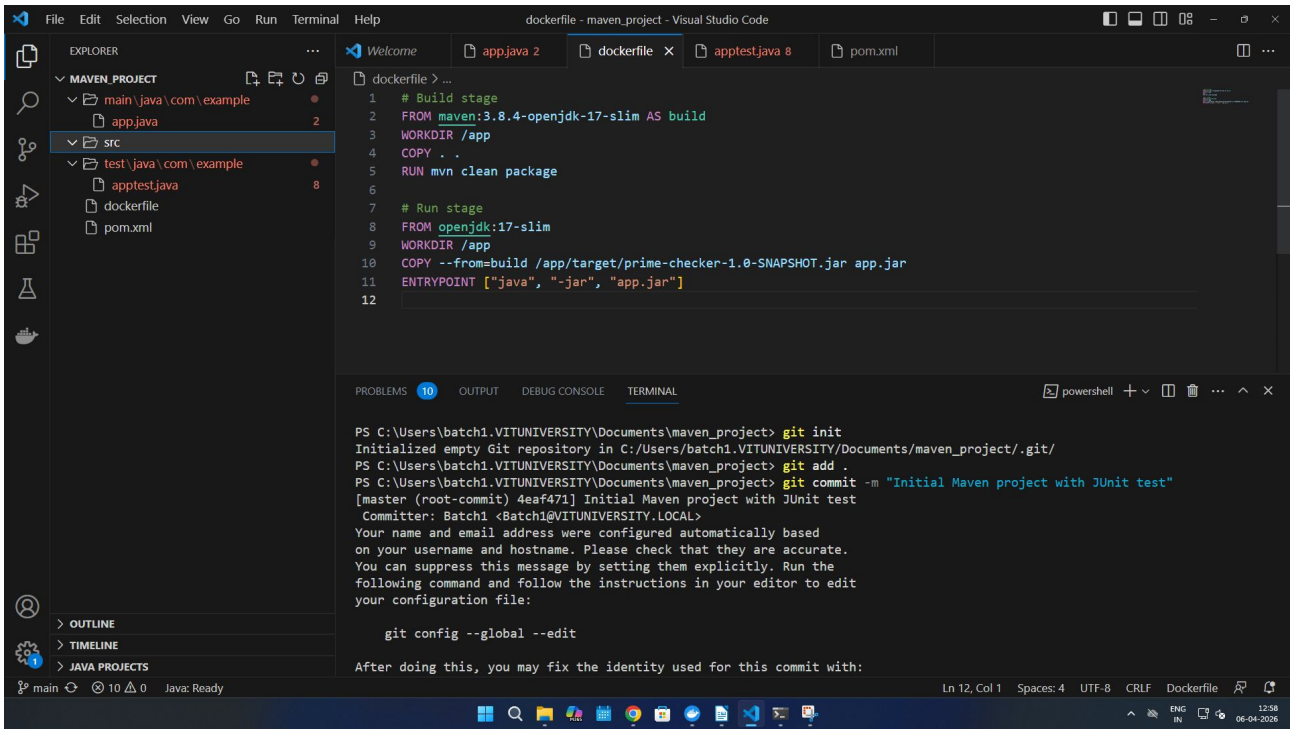
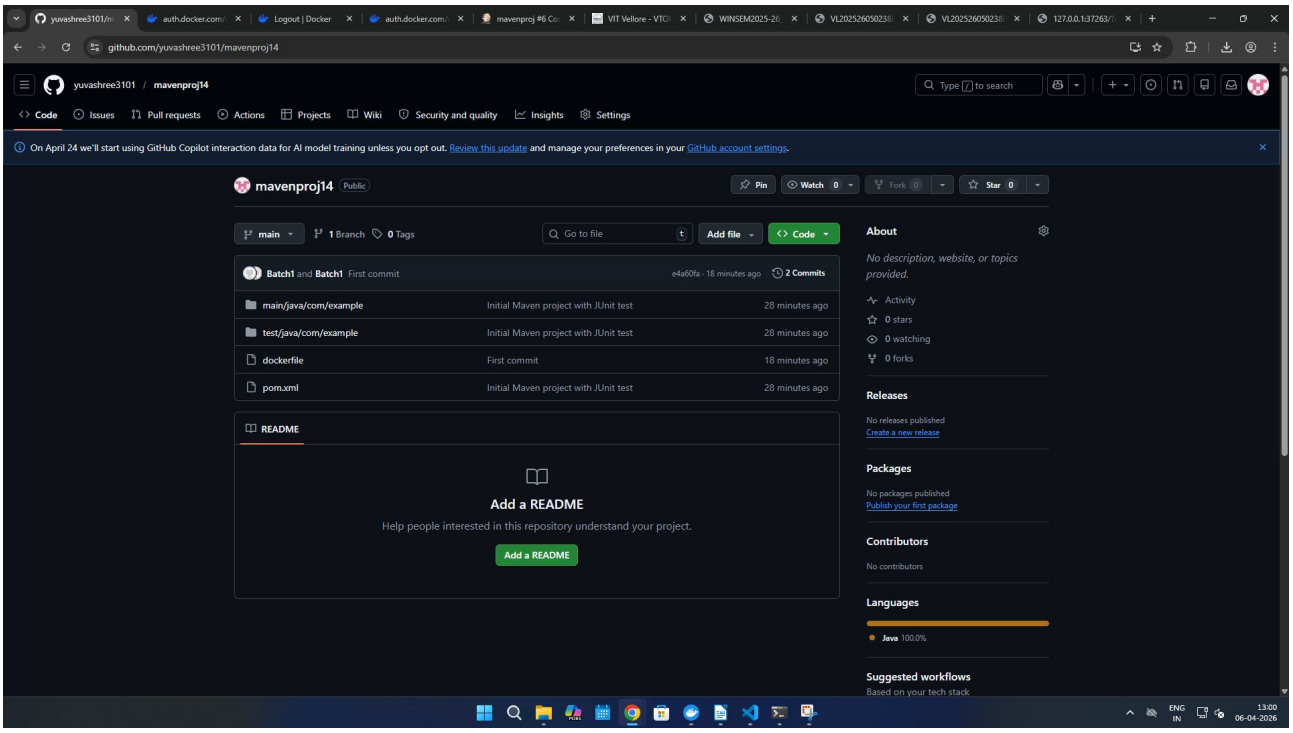
REG NO:23MIS0593

1. Simple Java Console App (Baseline CI/CD) TASK :
Create Maven project with Junit tests
Create a Maven based application by following standard Maven directory structure.
Write the code in Visual Studio and push it to your GITHUB repository.
Modify the code and commit the updates back in GITHUB.
Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.
Containerize the application using Dockerfile and display the stages in stage view in Jenkins.

e. Prime number.

SOLUTION:


dockerfile - maven_project - Visual Studio Code
EXPLORER
MAVEN_PROJECT
main\java\com\example
app.java
src
test\java\com\example
apptest.java
dockerfile
pom.xml
Build stage
FROM maven:3.8.4-openjdk-17-slim AS build
WORKDIR /app
COPY . .
RUN mvn clean package
Run stage
FROM openjdk:17-slim
WORKDIR /app
COPY --from=build /app/target/prime-checker-1.0-SNAPSHOT.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
PROBLEMS 10
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:\Users\batch1.VITUNIVERSITY\Documents\maven_project> git init
Initialized empty Git repository in C:/Users/batch1.VITUNIVERSITY/Documents/maven_project/.git/
PS C:\Users\batch1.VITUNIVERSITY\Documents\maven_project> git add .
PS C:\Users\batch1.VITUNIVERSITY\Documents\maven_project> git commit -m "Initial Maven project with JUnit test"
[master (root-commit) 4eaf471] Initial Maven project with JUnit test
Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:
git config --global --edit
After doing this, you may fix the identity used for this commit with:


github.com/yuvashree3101/mavenproj14
yuvashree3101 / mavenproj14
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
mavenproj14 Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Batch1 and Batch1 First commit
e4a60fa · 18 minutes ago
2 Commits
main/java/com/example
Initial Maven project with JUnit test
28 minutes ago
test/java/com/example
Initial Maven project with JUnit test
28 minutes ago
dockerfile
First commit
18 minutes ago
pom.xml
Initial Maven project with JUnit test
28 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors
No contributors
Languages
Java 100.0%
Suggested workflows
Based on your tech stack

CODE:

app.java

```
package com.example;

public class App {
    public static boolean isPrime(int n) {
        if (n <= 1) return false;
        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0) return false;
        }
        return true;
    }
    public static void main(String[] args) {
        int num = 17;
        System.out.println(num + " is prime? " +
isPrime(num));
    }
}
```

apptest.java

```
package com.example;
import static org.junit.Assert.*;
import org.junit.Test;

public class AppTest {
    @Test
```

```
    public void testIsPrime() {
        assertTrue(App.isPrime(7));
        assertFalse(App.isPrime(4));
        assertFalse(App.isPrime(1));
    }
}
```

dockerfile:

```
# Build stage
FROM maven:3.8.4-openjdk-17-slim AS build
WORKDIR /app
COPY . .
RUN mvn clean package

# Run stage
FROM openjdk:17-slim
WORKDIR /app
COPY --from=build /app/target/prime-checker-1.0-
SNAPSHOT.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
```

pom.xml:

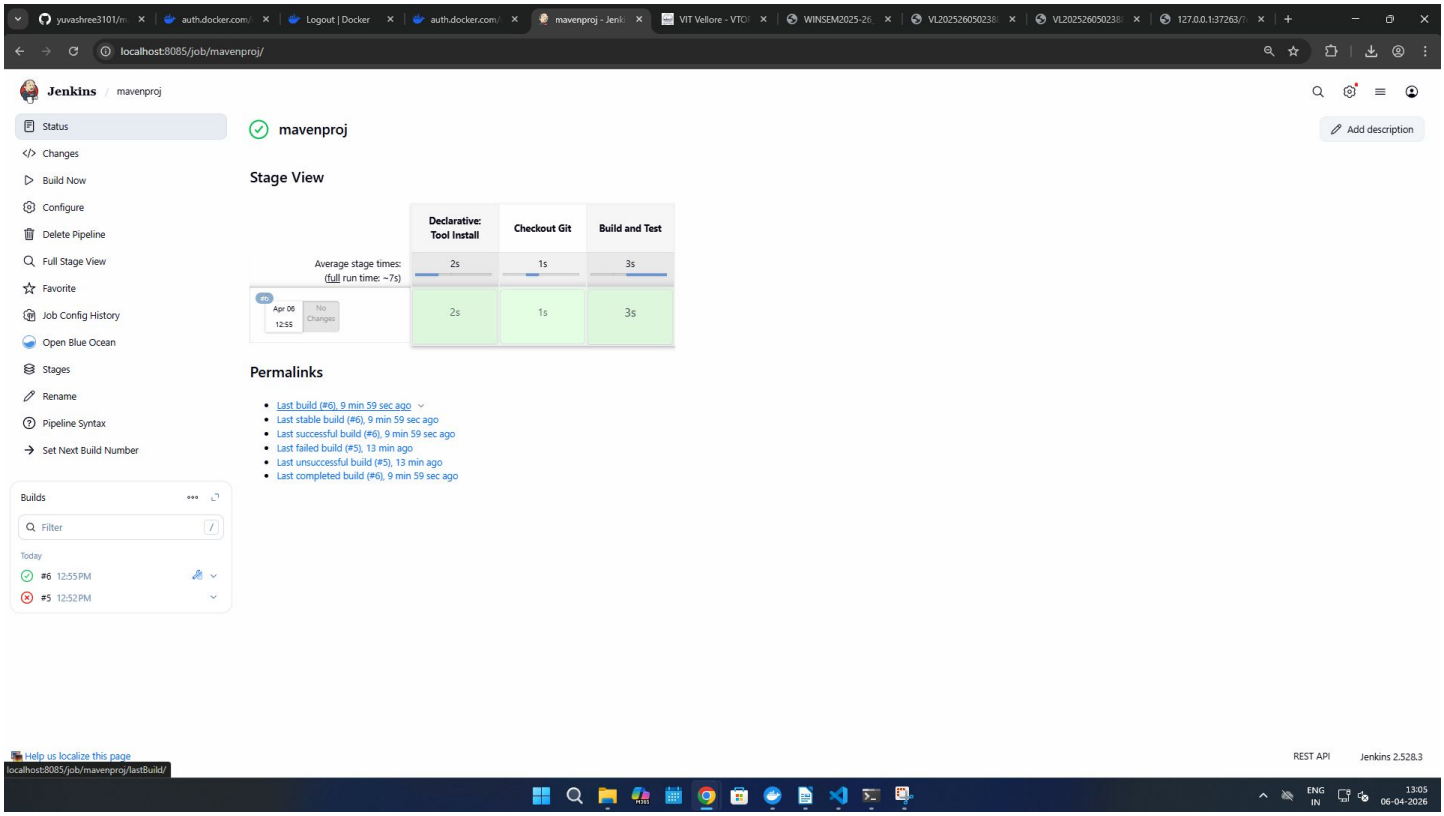
```
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.example</groupId>
 <artifactId>simple-maven-app</artifactId>
```

```
<version>1.0-SNAPSHOT</version>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>4.13.2</version>
<scope>test</scope>
</dependency>
</dependencies>
</project>
```

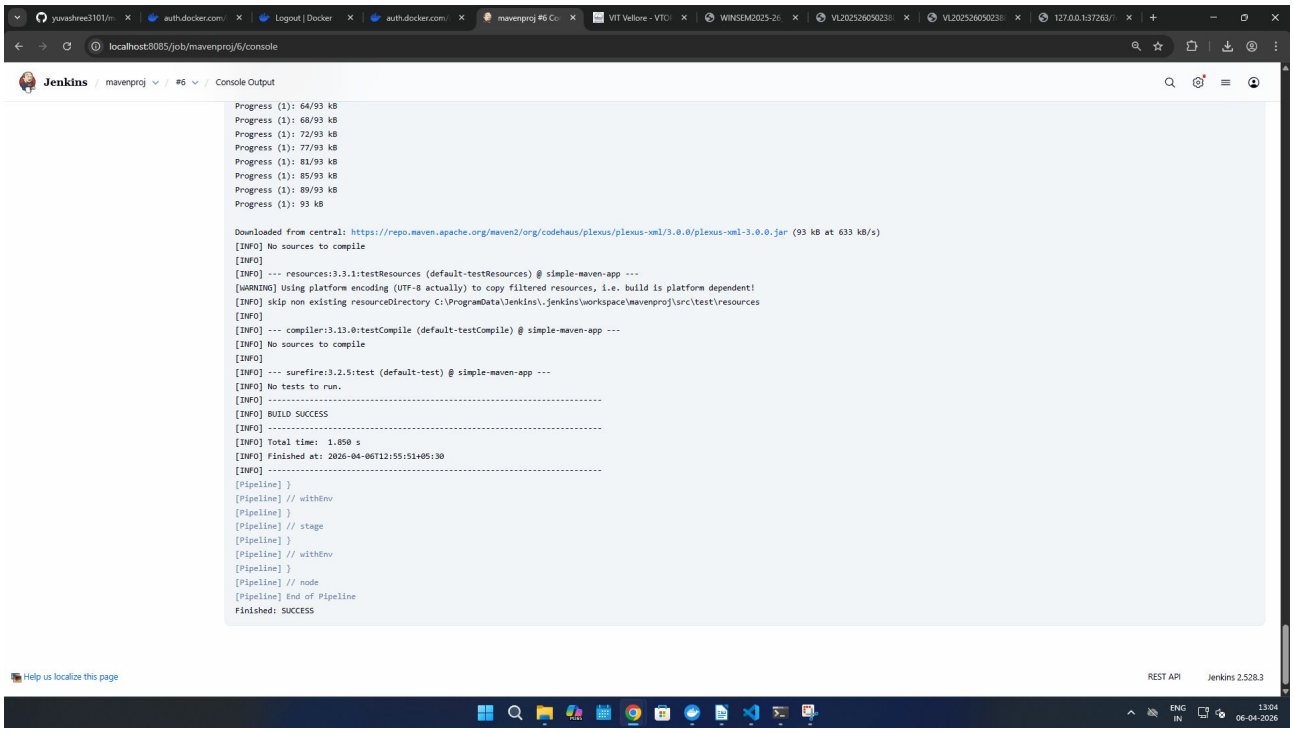
jenkins (pipeline):

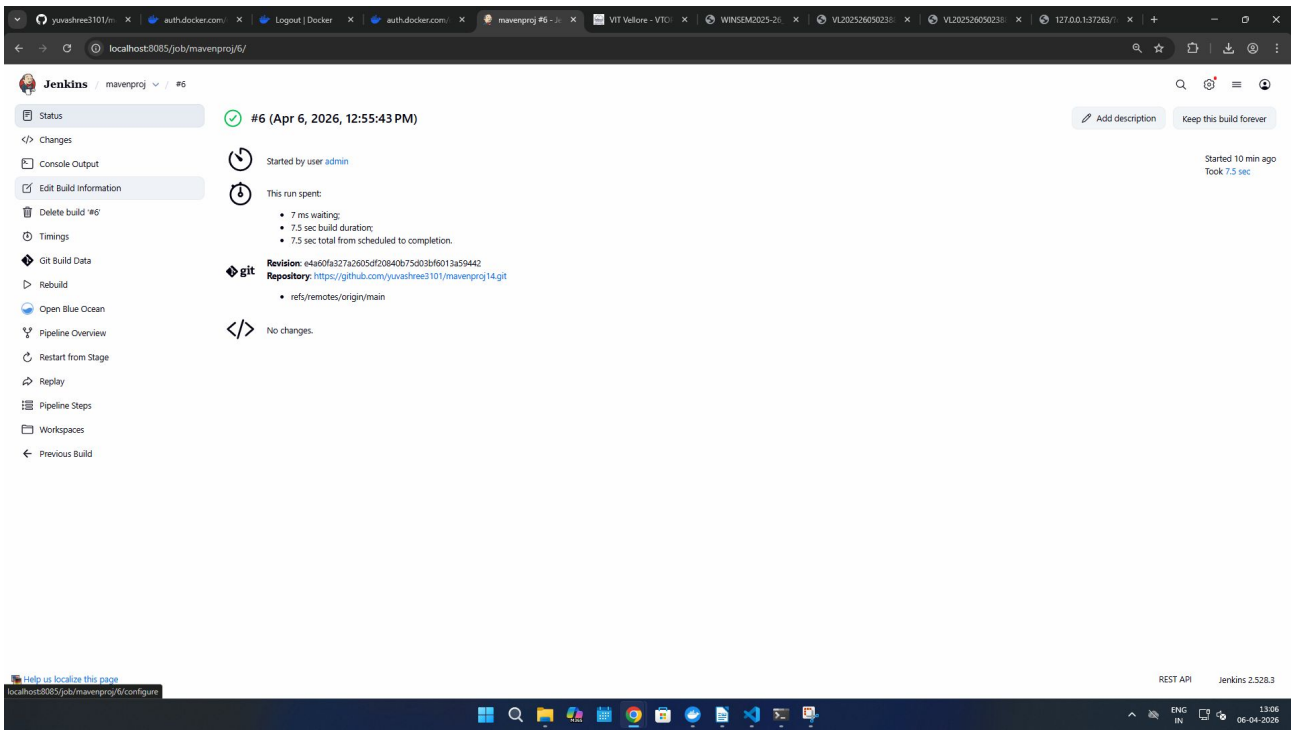
```
pipeline {
 agent any
 tools {
 maven 'M3'
 }
 stages {
 stage('Checkout Git') {
```

```
steps {
git branch: 'main',
url: 'https://github.com/yuvashree3101/mavenproj14.git'
}
}
stage('Build and Test') {
steps {
bat 'mvn clean test'
}
}
}
}
```



Console output:





console output:

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-xml/3.0.0/plexus-xml-3.0.0.jar (93 kB at 633 kB/s)
[INFO] No sources to compile

[INFO]

[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---

[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!

[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\mavenproj\src\test\resources

[INFO]

[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ simple-maven-app ---

[INFO] No sources to compile

[INFO]

[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---

[INFO] No tests to run.

[INFO] ------------------------------------------------------------------------

[INFO] BUILD SUCCESS

[INFO] ------------------------------------------------------------------------

[INFO] Total time:  1.850 s
```

```
[INFO] Finished at: 2026-04-06T12:55:51+05:30

[INFO] ------------------------------------------------------------------------

[Pipeline] }

[Pipeline] // withEnv

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // withEnv

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS
```